Additive Structure of Phonological Correspondences

A protoform-agnostic method for discovering mathematical patterns in documented linguistic systems

Alejandro Toledo Martínez — Independent researcher (ORCID: 0009-0000-1277-9697)

Abstract

Historical linguistics infers ancestral forms and then derives documented ones from them. We invert that order. Representing each attested sound correspondence not as a directed change but as the set of phonological features that differ between two aligned segments — a symmetric *operator* — a language family becomes a repertoire O of feature-difference vectors over $\mathbb{F}_2$, computed with no reconstruction, no sound laws, and no externally imposed sound classes. We study the geometry and additive structure of O for Indo-European (IE) and Austronesian (AN), drawn reconstruction-free from Lexibank via statistical cognacy (LexStat) and feature-distance alignment (panphon). Four results. (i) Occupancy is not sparsity of the languages but vastness of the space: of the differences the corpus actually makes available, IE realizes 63% and AN 48%. (ii) The repertoire is additively organized — combining two operators lands back in O far more often than chance, and this survives a hierarchy of six nested null models up to and including sampling from the corpus's own opportunity set ($Z = +6.3$ IE, $+5.9$ AN; $p_{\mathrm{MC}} \leq 0.004$). (iii) That additive structure is a property of the phonological representation, not of genealogy: a concept-permuted control matches it, and branch repertoires overlap no more than random language groupings in IE (though they do in AN). (iv) The generable-but-unattested operators $\langle O \rangle \setminus O$ form a concrete geometry of near-misses. The contribution is not a sound law but an *object of study* — the empirical repertoire of correspondences and its measurable additive geometry — together with an inference order that keeps the historical apparatus out of the computation so it can serve as an independent test. A companion manual gives the full didactic treatment.

1. Introduction

The comparative method is an inference toward a hidden past: from documented forms X it reconstructs an ancestor $\widehat{Z}$ and posits the changes H that produced X. It answers *how did these forms descend from their common ancestor?* We ask a different question — *treated purely as a set of differences, what shape does the space of attested correspondences have?* — and we answer it in the opposite order:

$$X \longrightarrow \mathcal{M}(X), \qquad \text{then, separately,} \qquad \mathcal{M}(X) \longleftrightarrow H,$$

where $\mathcal{M}(X)$ is a mathematical structure discovered without protoforms or precoded laws, and the historical apparatus enters only *afterward*, as contrast. This is not a rejection of reconstruction; it is a more demanding test of it. Protoforms are inferred *from* the documented languages; building the space *on* them and then checking whether it confirms historical regularities risks circularity — leakage of the target into the inputs. Suspending them during discovery is therefore a design choice, not a limitation.

Two clarifications frame everything below. First, the method is not "theory-free": transcription, segmentation, concept assignment, the feature matrix, and the aligner are measurement instruments whose influence we make explicit and control for (§3, §4.3). Second, this is emphatically not a claim that "sound change is XOR," nor a hunt for known sound laws; it is a change of what is measured and in what order.

2. The operator and the repertoire

For two aligned segments a, b with panphon feature map ϕ, the operator is the symmetric feature-difference

$$\Delta(a,b) = \{k : \phi_k(a) \neq \phi_k(b)\} \;=\; \phi(a) \oplus \phi(b) \;\; \text{(over binary features)}.$$

It is a *contrast*, not a directed change: $\Delta(a,b) = \Delta(b,a)$, so Δ does not distinguish $t \to s$ from $s \to t$, and "each operator is its own inverse" is a consequence of that symmetry, not a phonological finding. Directed history, when wanted, requires a richer object $T = (a, b, c, p, \ell_1, \ell_2, w)$ with signature $\sigma(T) = \Delta(a,b)$; this paper studies the symmetric signatures. We keep operators of one to three features (atoms and small molecules).

A family's repertoire O is its set of recurrent operators. O generates a subspace $\langle O\rangle = \mathrm{span}_{\mathbb{F}_2}(O)$ of dimension (rank) r; crucially O is not a subspace (not closed under XOR, does not contain 0) — it is a *region*, and the object of interest is the relation between the region O and the capacity $\langle O\rangle$.

3. Data and method

Corpus. Lexibank word lists, IPA-segmented, for Indo-European (304 languages available) and Austronesian (978), taking per family the languages with the most forms. Cognacy is detected statistically with LexStat (LingPy); alignment is Needleman–Wunsch with substitution cost equal to the fraction of panphon features that differ; features are panphon's primary set (12 features including labiality). No protoforms, laws, or sound classes enter the computation. Operators are taken from aligned, non-identical, consonantal positions (vowels are ablaut noise) with support thresholds (30 at family scale). Independent validation uses IE-CoR expert cognacy (§4.6). Every quantity is reproducible from a single `make` target (§7).

Two families give a minimal comparative frame; "IE" and "AN" are pilot genealogical boundaries, not results.

4. Results

4.1 Occupancy: sparsity is mostly algebraic

IE uses $|O| = 67$ operators, AN $|O| = 22$, with ranks $r = 12$ and $r = 10$. Against the full algebraic span these are 1.6% and 2.2% — but the span $2^r - 1$ contains feature-bundles that no pair of real segments could realize. Nested between O and $\langle O\rangle$ are the realizable universe $U_S = \{\Delta(a,b) : a, b \in S\}$ (over the attested segment inventory S) and the opportunity universe Ω_D (differences that actually occur in aligned corpus positions):

family	ρ_{alg} (vs span)	ρ_{seg} (vs realizable)	ρ_{opp} (vs opportunity)
Indo-European	0.016	0.48	0.63
Austronesian	0.022	0.43	0.48

where $\rho_{\text{alg}} = |O|/(2^r - 1)$, $\rho_{\text{seg}} = |O|/|U_S \cap \langle O \rangle|$, and $\rho_{\text{opp}} = |O|/|\Omega_D \cap \langle O \rangle|$. Of what was actually available, the repertoire is *dense*, not sparse: the "98% unused" is the algebraic space being enormous, not languages being restrictive.

4.2 The repertoire is additively organized

Composing two operators is XOR of their feature-sets. The composition-realization index

$$C(O) = \frac{|\{\{u, v\} \subseteq O : u \oplus v \in O\}|}{\binom{|O|}{2}}$$

is 0.220 (IE) and 0.234 (AN); conditioned on the opportunity universe, $C_\Omega(O) = 0.81$ (IE), 0.66 (AN). To test whether this exceeds chance we compare against a hierarchy of null repertoires, each preserving more structure and each sampled to size $|O|$; we report Z and empirical p_{MC} over 500 simulations:

null preserves	IE Z	AN Z
size + weight ≤ 3	+15.7	+12.0
Hamming weights	+14.8	+9.0
exact margins (swap-MCMC)	+7.5	+3.6
rank / span $\langle O \rangle$	+45.6	+13.7
realizable U_S	+9.2	+7.3
opportunity Ω_D	+6.3	+5.9

All six survive at $p_{\text{MC}} \leq 0.004$: even drawn from exactly the differences the corpus made available, the observed repertoire is significantly more composition-closed. Three further additive statistics agree (against the Ω_D null): triple density τ is high (IE Z=+6.6, AN $+6.5$), the doubling constant $\kappa = |O \oplus O|/|O|$ is low (IE Z=-5.1, AN -6.0), and additive energy E is high (IE $+7.4$, AN $+7.3$). The repertoire behaves like an approximate (possibly affine) subspace. Its dependency structure has a basis-independent core: as a binary matroid it has 162 (IE) and 18 (AN) three-element circuits, with `{voi}` (voicing) the most-connected operator in both — the additive hub.

4.3 What the additive structure is — and is not

The structure of §4.2 is a property of the phonological representation and inventory, not of genealogy or semantics. Two controls establish this. First, permuting concept labels to destroy all cognacy and semantic linkage (regime D_R) yields additive structure *as high as or higher than* expert cognacy D_G (C=0.215 vs 0.169), so the type-level additive closure does not distinguish history from inventory. Second, the one linear dependency we observe — in AN, `ant` = `cor`$\oplus$`lab` — is a property of the *selected* repertoire, not of the feature matrix: the realizable universe U_S has full rank (12, corank 0), so the ontology would permit independence; the dependency appears only in O_{AN}. The additive geometry is real (§4.2) but must be read as *representational*; genealogical signal lives elsewhere (§4.5–4.6).

4.4 The geometry of holes $\langle O \rangle \setminus O$

Among the operator-shaped (weight ≤ 3) vectors the span generates, most are holes — generable but unattested. For each hole x, its depth is $d(x, O) = \min_{o \in O} d_H(x, o)$.

	generable (weight ≤ 3)	observed	holes	at $d = 1$	holes in Ω_D
Indo-European	298	67	231	160	40
Austronesian	119	22	97	53	24

Most holes lie one feature from an attested operator; the linguistically loaded ones are the holes in Ω_D (40 IE, 24 AN) — differences the corpus made available yet the system left unused. These are the concrete candidates for interpretation: absences that are choices, not impossibilities.

4.5 Systems and branches

Partitioning each family into Glottolog branches and taking $O_F = \bigcup_r O_r$, the persistence histogram H_k (operators in exactly k branches) decomposes the repertoire without double-counting: IE $|O_F| = 97$, strict nucleus $K_\cap = 12$ (the atoms `{ant}` `{cont}` `{voi}` in every branch), majority nucleus $K_{1/2} = 38$; AN $|O_F| = 48$, $K_\cap = 4$, $K_{1/2} = 18$. But raw branch overlap is uninterpretable without a null. Against a grouping null (arbitrary same-size language groups) the mean Jaccard tells opposite stories in the two families:

	mean Jaccard (real branches)	arbitrary groupings	Z
Indo-European	0.42	0.45 ± 0.05	-0.7 (indistinguishable)
Austronesian	0.51	0.57 ± 0.01	-5.4 (real branches more differentiated)

At the level of operator types, Indo-European branches share no more than random groupings — the apparent "shared core" is shared *inventory* — whereas Austronesian branches are significantly more differentiated than chance. Under rarefaction the nucleus size is sample-sensitive, so branch *richness* is not interpreted; the robust facts are the histogram decomposition and the grouping-null contrast.

4.6 Validation against expert cognacy

Statistical cognacy could contaminate the repertoire with false cognates. Against IE-CoR expert cognacy the contamination is real at the instance level (pair precision 0.73, recall 0.33) but absorbed at the type level: LexStat's operator inventory has precision ≈ 1.0 and recall 0.66 against the expert repertoire — a false cognate almost always produces an operator already present. The type inventory is thus more robust than individual cognate identification. Where genealogical signal does live is the distributions: mutual information between operator and branch is modest but non-zero and larger in AN than IE (0.22 vs 0.14 normalized), consistent with §4.5; a few operators dominate each family (effective count $N_{\text{eff}} = 24$ of 119 IE, 18 of 63 AN).

5. Figures

6. Discussion

The contribution is an object and an order. The object is the empirical repertoire O of symmetric feature-difference operators, and the separation between it and the subspace $\langle O \rangle$ it generates: a system uses a small, additively-structured region of a large algebraic space, densely relative to opportunity, with a concrete geometry of unused near-misses. The order is discover-then-contrast, which

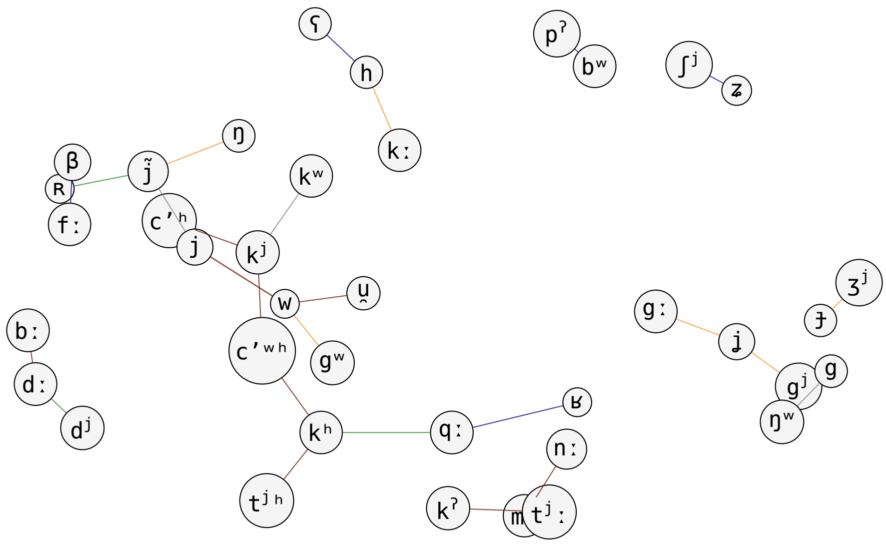

Figure 1: Segmental correspondence graph G_S of Indo-European. Nodes are segments; edges are recurrent correspondences, with thickness $\propto$ frequency and colour by change type: voicing, fricativization/lenition, palatalization/height, place, prenasalization.

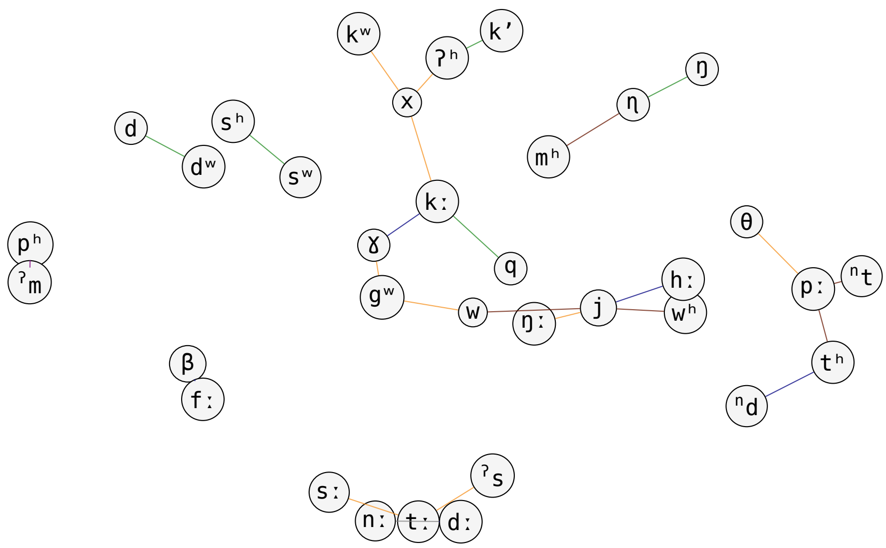

Figure 2: Segmental correspondence graph G_S of Austronesian. Nodes are segments; edges are recurrent correspondences, with thickness $\propto$ frequency and colour by change type: voicing, fricativization/lenition, palatalization/height, place, prenasalization.

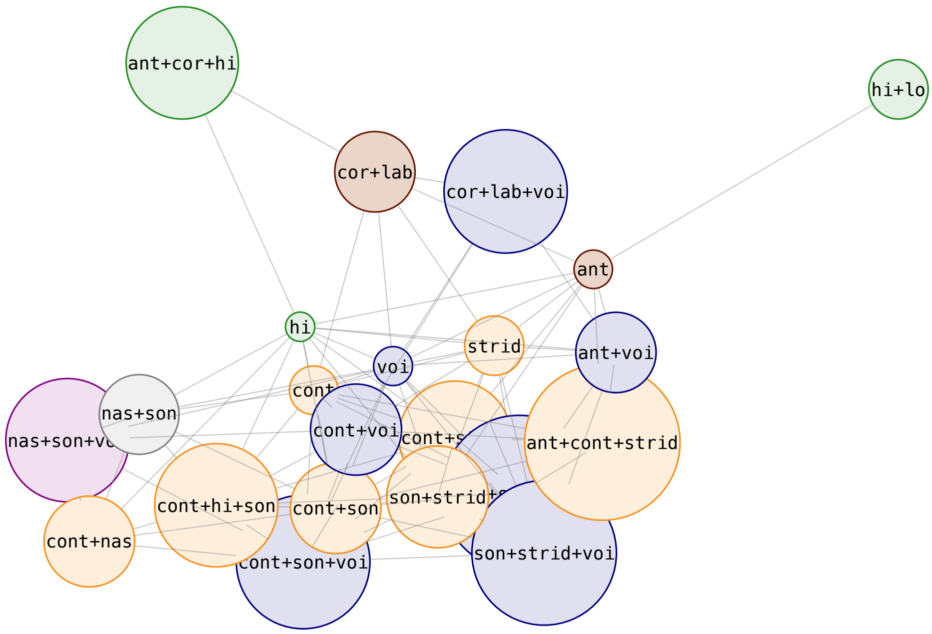

Figure 3: Operator graph G_O of Indo-European. Nodes are operators (the 22 with highest support), sized by frequency and coloured by change type (voicing, fricativization, palatalization/height, place, prenasalization). An edge joins u, v when their composition $u \oplus v$ is itself an operator of the repertoire — so this is the graph of *additive closure*. Dense connectivity is the visual face of the composition index $C(O)$.

keeps reconstructions, laws, and sound classes out of the measurement so each can serve as an independent yardstick later.

Two boundaries keep the claims honest. The additive structure is *representational*, not genealogical (§4.3); the genealogical question is properly posed on distributions and, in future work, on a directed, context-conditioned layer — the tensor X_{ijoc} of operator frequency by language pair and context, whose non-negative factorization and minimum-description-length rules would let sound laws *emerge* as compressive bundles rather than be inserted. Under a historical-holdout protocol (discover; test stability; cluster languages blind; only then compare to reconstructions and trees), a recovered law would be evidence precisely because it was never an input. The same discipline — discover structure first, admit ground-truth labels only to test — is a template for leakage-free evaluation of language models on historical and typological tasks.

7. Limitations and reproducibility

Limitations. Two families are a pilot; "family" is a genealogical label taken as a boundary, not a result. The opportunity universe Ω_D and the additive nulls depend on the corpus and aligner. The additive structure is type-level and representational; loans-as-system-influence could not be tested (loan annotations are absent in the pilot corpora and require a resource such as WOLD). Force-directed figures lack an articulatory coordinate system. None of these affect the null-controlled results, but they bound their interpretation.

Reproducibility. All results run from `transformations/src` against a repertoire built by `make family FAMILY="X"`. Key targets: `make universes` (§4.1), `make nulls` / `make additive` (§4.2, §4.4), `make repr-control` (§4.3), `make superposition` (§4.5), `make cognate-eval` / `make distributions` (§4.6). Datasets: Lexibank, IE-CoR, Glottolog (cached classification). LexStat threshold 0.55, 100-run scorer; panphon 12-feature primary set; seeds fixed. The companion manual (`docs/manual`) gives worked, hand-computed examples of every measure and the full derivations.